The Dot Product

Computing dot products and angles

1 Let $\mathbf{u} = \langle 2, 5 \rangle$ and $\mathbf{v} = \langle -3, 4 \rangle$. Find:

a $\mathbf{u} \cdot \mathbf{v}$

b $|\mathbf{u}|$ and $|\mathbf{v}|$

c the angle between $\mathbf{u}$ and $\mathbf{v}$, to one decimal place

2 Determine whether each pair of vectors is orthogonal:

a $\langle 4, -6 \rangle$ and $\langle 3, 2 \rangle$

b $\langle 1, 2 \rangle$ and $\langle 2, 1 \rangle$

3 Find the value of k so that $\langle k, 3 \rangle$ is orthogonal to $\langle 2, -4 \rangle$.

Projections and work

4 Find the vector projection of $\mathbf{u} = \langle 6, 2 \rangle$ onto $\mathbf{v} = \langle 3, 0 \rangle$.

5 A force $\mathbf{F} = \langle 8, 5 \rangle$ N moves an object along the displacement $\mathbf{d} = \langle 10, 0 \rangle$ m. Find the work done.